

Yingrui Hou

Industrial Data Scientist - Statistical Modelling, Validation and Monitoring

Clermont-Ferrand, France | +33 (0)785396332 | yingrui.hou@outlook.com

LinkedIn: [linkedin.com/in/yingrui-hou](https://www.linkedin.com/in/yingrui-hou) | GitLab: gitlab.com/hyrsophie | CERN: gitlab.cern.ch/yihou

SUMMARY

R&D-oriented simulation and data scientist with a background in experimental physics and several years of experience building reusable Python/C++ workflows for calibration, validation, and inference on noisy measurement systems.

Core strength: turning imperfect simulations and biased measurements into decision-ready models through simulation-to-data alignment, correction workflows, uncertainty handling, and robust validation.

Best fit: industrial R&D teams working on physical systems, instrumented tests, sensor data, model credibility, and simulation-informed engineering decisions.

CORE SKILLS

- Machine learning: CatBoost / Gradient Boosting, Domain Adaptation
- Data analysis stack: NumPy, pandas, scikit-learn, statsmodels, matplotlib / seaborn
- Statistical modelling: Probabilistic Modelling, Statistical Inference, Maximum Likelihood Estimation, Signal Processing
- Validation and monitoring: Model Validation, Calibration & Bias Correction, Simulation-to-Data Alignment, Large-Scale Data Analysis
- Engineering workflow: Python, C/C++, ROOT / RooFit, Workflow Automation (YAML / Shell / CMake)

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Built reusable C++/Python/ROOT workflows for weak-signal extraction, time-dependent inference, and validation on noisy datasets where direct event-level truth was unavailable.
- Implemented likelihood-based models and joint fits to estimate latent parameters under sample mismatch, finite resolution, and biased observation conditions.
- Developed CatBoost-based selection and validation pipelines, including train/test diagnostics, scan studies, and reproducible model application.
- Aligned simulated and observed datasets through gradient-boosting reweighting and control-sample calibration before downstream inference and selection decisions.
- Contributed reusable fit components and uncertainty-stress checks for advanced weighted and unbinned inference workflows.
- Worked in large technical collaborations with shared codebases and review discipline, with validation outputs feeding calibration and model decisions.
- Developed ROOT/C++ calibration and correction workflows to reduce structured bias and compare performance before and after correction across operating conditions.
- Built benchmark-based monitoring workflows to validate reconstruction and calibration quality, then traced anomalies to detector behaviour or software issues instead of stopping at descriptive analysis.
- Identified concrete issues that led to follow-up actions, including detector-specific anomalies and a missing-energy software problem later fixed in reconstruction software.

SELECTED OUTCOMES

- Quantified response non-uniformity contribution to resolution below 1% in a detailed detector study
- Improved reconstructed-energy resolution by about 2x in representative PSD-based studies
- Built validation workflows that exposed detector and software issues, including a missing-energy bug later fixed downstream

TECHNICAL HIGHLIGHTS

- Model validation and uncertainty quantification for noisy measurement systems
- Simulation-to-measurement alignment through calibration, reweighting, and correction workflows
- Sensor and system modelling with Geant4, ROOT, Python, and C++
- Test-data analytics, anomaly tracing, and root-cause diagnosis
- ML validation under dataset shift for decision-facing pipelines

EDUCATION

- PhD in Particle Physics - University of Chinese Academy of Sciences
- BSc in Applied Physics - China University of Mining and Technology
- Languages: Chinese: Native speaker, English: C1, French: B1 to B2